

An Abstract Drawing Method for Same Shaped but Densely Arranged Many Objects

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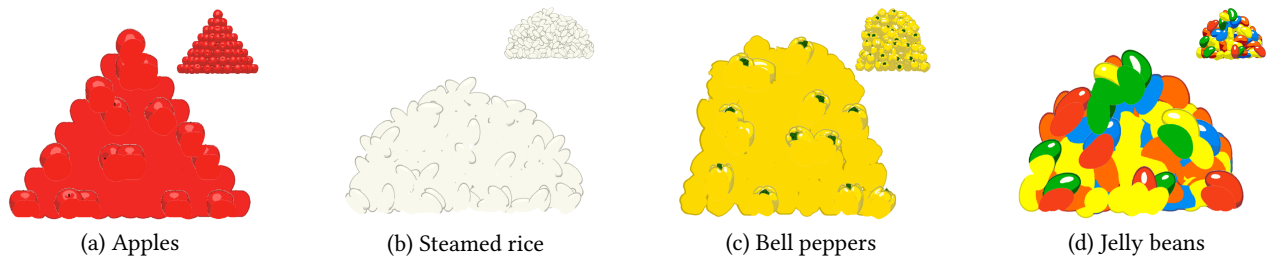


Figure 1: Abstraction results for densely arranged objects (Each top right image shows a result of a conventional toon rendering).

ABSTRACT

We propose a hierarchical abstract drawing method for many 3D objects which are densely arranged and have almost the same shape. First, an abstract drawing of all the 3D objects is drawn based on a couple of the primal colors, which is one of the global properties of all the 3D objects. Then, an abstract painting is added focusing on a part of each object as local properties. Several results of abstractly rendered scenes are shown, and are confirmed the effectiveness of our method.

CCS CONCEPTS

• Computing methodologies → Non-photorealistic rendering.

KEYWORDS

non-photorealistic rendering, real-time abstraction, densely arranged objects

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1 INTRODUCTION

Abstraction is one of the most important non-photorealistic expressions in manga and animation. In manga and animation, we often see scenes where 3D objects of similar shape are densely arranged. It would be difficult to draw thousands of 3D objects of almost the same shape and colors one by one. It takes a lot of time to draw such a picture. Therefore, in manga and animation, the scenes are represented by mixing the flatly painted part and the faithfully painted part of each object. It would be applicable to use the levels of detail technique (LoD), however, drawing dense objects with the LoD technique often loses details.

We propose a method of drawing many 3D objects by a two-stage abstraction. The first stage is a rough abstraction stage that outlines a heap of many densely arranged 3D objects. Then a combination of a toon shader and flat color representing the 3D objects is utilized to paint the inside of the contour. The second stage is a fine abstraction stage, in which a part of specific objects is abstracted from among the 3D objects heaped up. Our method combines rough and fine abstract drawing techniques to effectively represent a scene in which many 3D objects of almost the same shape and colors are densely arranged (Figure 1).

2 RELATED WORK

There have been proposed a couple of methods for generating tree illustrations. Deussen and Strothotte used the Z-buffer to abstract 3D objects of leaves from a 3D tree object and generated an illustration of the tree [Deussen and Strothotte 2000]. Di Fiore et al. combined an illustration of a group of leaves pre-drawn by the user with an illustration of only the trunk and branches drawn from a 3D tree object [Di Fiore et al. 2003]. These methods are dedicated to tree abstraction. Therefore, it is not suitable for abstracting scenes where objects with structures other than trees are densely arranged.

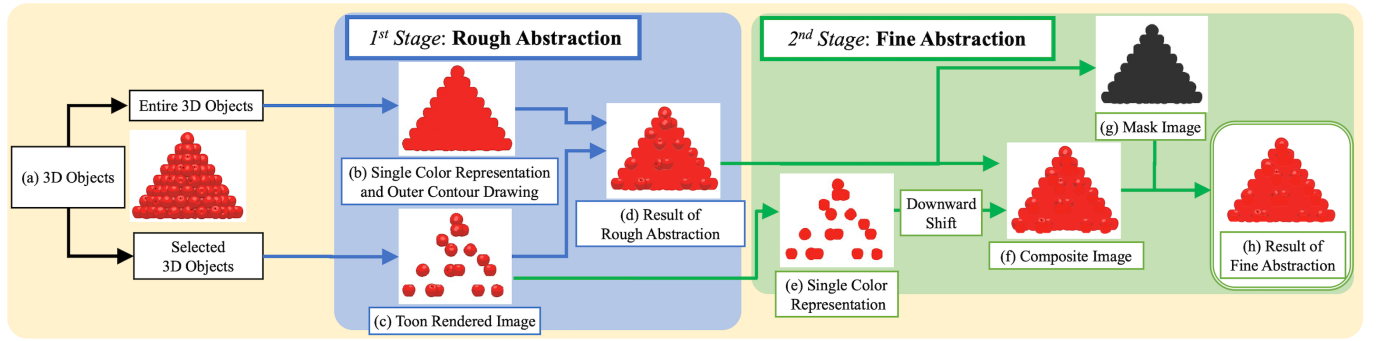


Figure 2: An overview of the proposed method.

3 PROPOSED METHOD

3.1 Overview

Figure 2 shows an overview of our proposed method. We use a 3D object of an apple in this case because it has simple color. In the proposed method, a two-stage abstract representation of densely arranged objects is realized, such as rough abstraction (Section 3.2) and fine abstraction (Section 3.3).

3.2 Rough Abstraction

We manually densely arrange entire 3D objects (Figure 2(a)). By applying toon shading to the 3D objects, we obtain an image with reduced colors.

We change the color of the 3D object to a single color. In this case, we use red. For example, we can use the color for the most frequent color among the colors in Figure 2(a). We outline the entire 3D objects (Figure 2(b)). We manually select non-abstracted 3D objects from entire 3D objects and draw only the selected 3D objects (Figure 2(c)). In this case, we select a quarter of 3D objects. Finally, by synthesizing Figure 2(b) and Figure 2(c) with depth comparison, we obtain a rough abstraction result (Figure 2(d)).

3.3 Fine Abstraction

In this stage, we further abstract the regions that were not abstracted in the previous stage.

First, we convert Figure 2(c) to a single color (Figure 2(e)). We shift Figure 2(e) downward and synthesize it into Figure 2(d) with depth comparison to abstract details (Figure 2(f)). By using a mask image (Figure 2(g)) generated from Figure 2(d), we prevent the result (Figure 2(h)) from being drawn outside the arranged objects.

4 IMPLEMENTATION AND RESULTS

We implemented the proposed method into the deferred rendering pipeline. The proposed method can run in real-time.

4.1 Various Densely Arranged Objects

Figure 1 shows the results using various inputs. Figure 1(a) shows a drawing result of many apples arranged in a pyramidal style. Figure 1(b) shows a drawing result of steamed rice. Each 3D object is arranged by a hemispherical shape. Figure 1(c) shows a drawing result of bell peppers arranged in a conical shape. Figure 1(d) shows

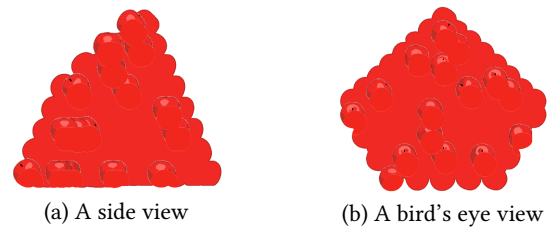


Figure 3: Rendering results from different viewpoints.

a drawing result of jelly beans, where jellybeans have different colors (five colors in this case). The 3D objects are arranged in a hemispherical shape.

From these drawing results, it is confirmed that the proposed method effectively enables artistic and abstract expressions regardless of the materials or properties of 3D objects such as shapes and colors.

4.2 Rendering from Different Viewpoints

Our method allows us to abstract and represent densely arranged objects rendered from different viewpoints. Figure 3 shows the results rendered from different viewpoints of Figure 1(a).

5 CONCLUSION

We proposed a method to abstract and render many 3D objects that are densely arranged with almost the same shape. It is confirmed that our method is effective since it became possible to apply our proposed method to various 3D objects, and abstract and render them from different viewpoints.

In the future, we would like to realize several functions such as automatically arranging 3D objects as well as automatically selecting objects and areas to be abstracted. We will also tackle temporal consistency and changes in appearance according to distance for future improvement.

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